

QIUSHI LIN

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EDUCATION

The University of Texas at Austin

2026—Present

Ph.D. Student in Computer Science

Advisor: [Prof. Jovan Stojković](#)

Tsinghua University

Sep. 2022—2026

B.S. in Electronic Engineering

GPA: 3.9/4.0 (Graduated with Honor)

GPA Rank: 3/263

PUBLICATIONS

1. **AtumAI: A Principled Framework for Agentic Generation of Datacenter Control-Plane Policies** ([Paper](#))
Qiushi Lin, Chaojie Zhang, Íñigo Goiri, Aditya Akella, Ricardo Bianchini, and Jovan Stojkovic.
Preprint.
2. **Gutenberg: Taming Latency-Critical Cloud Services with Near-Data-Processing** ([Paper](#))
Qiushi Lin, Phillip B. Gibbons, Jovan Stojkovic, and Yiwei Zhao.
Preprint.
3. **TD-Orch: Scalable Load-Balancing for Distributed Systems with Applications to Graph Processing** ([Paper](#))
Qiushi Lin^{*}, Yiwei Zhao^{*}, Hongbo Kang, Guy E. Blelloch, Laxman Dhulipala, Charles McGuffey, and Phillip B. Gibbons.
Preprint.
4. **How Do Errors Impact NN Accuracy on Non-Ideal Analog PIM? Fast Evaluation via an Error-Injected Robustness Metric** ([Paper](#) | [Slides](#) | [Code](#))
Lidong Guo, Zhenhua Zhu⁺, **Qiushi Lin**, Yuan Xie, Huazhong Yang, Wangyang Fu⁺, and Yu Wang⁺.
ICCAD'25.
5. **HPIM-NoC: A Priori-Knowledge-Based Optimization Framework for Heterogeneous PIM-Based NoC** ([Paper](#) | [Code](#))
Shuai Yuan, Angxin Cai, **Qiushi Lin**, Guoxing Wang, Yu Wang, Zhenhua Zhu, and Yanan Sun.
DAC'25.
6. **Deep Neural Network Inference Partitioning in Embedded Analog-Digital Hybrid Systems** ([Paper](#))
Fabian Kreß, Julian Hoefer, **Qiushi Lin**, Patrick Schmidt, Zhenhua Zhu, Yu Zhu, Tanja Harbaum, Yu Wang, and Jurgen Becker.
ISQED'25.

SELECTED RESEARCH EXPERIENCE

Graduate Research Assistant, CASCADE Lab

2026—Present

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- Research on datacenter design, with a current focus on AI agents for cloud computing.

Task-Data Orchestration in Distributed Memory Systems

Nov. 2024—Oct. 2025

Carnegie Mellon University | *Advisor: Prof. Phillip B. Gibbons, Prof. Guy E. Blelloch*

- Designed a skew-aware load-balancing orchestration algorithm to optimize task-data allocation in distributed memory systems and built a graph processing engine on top of the framework.

- Formulated the theoretical foundations of the scheduling algorithm and architected the system core with $\sim 10,000$ lines of C++/MPI code, conducting extensive testing and optimization in distributed environments.
- Co-first-authored the TD-Orch preprint. The framework achieves up to $2.7\times$ speedup over distributed scheduling baselines, and the derived system TDO-GP achieves $4.1\times$ average speedup over prior state-of-the-art distributed graph processing frameworks.

Scheduling Algorithm and Microarchitecture Design for Processing Near Memory

Apr. 2025—2026

Carnegie Mellon University | Advisor: Prof. Phillip B. Gibbons, Prof. Guy E. Blelloch, and Prof. Yu Wang

- Conducted a comprehensive survey on DRAM-based processing-in-memory (PIM), covering hardware architectures, system-level optimizations, programming models, and security challenges, and summarized future directions for heterogeneous computing.
- Co-designed a QoS- and tail-latency-aware scheduling framework for PIM, addressing skewed access patterns, hot data regions, and multi-tenant workloads. Built a microservice-inspired execution model that jointly optimizes data placement, parallelism, and memory-access affinity across PIM modules.
- Developed a general-purpose PIM simulation framework that models a wide range of DRAM-PIM architectures and integrates host-CPU timing, enabling unified full-system evaluation of kernel latency, data movement, and QoS-aware scheduling policies across CPU and PIM cores.

NIPA: A Non-slicing Accuracy Evaluation Framework for Analog PIM (ICCAD'25)

Sep. 2024—Mar. 2025

Tsinghua University | Advisor: Prof. Yu Wang

- Contributed to the theoretical framework for unifying diverse hardware errors (device variation, quantization, ADC) into a single weight-level error model, enabling joint error analysis and bypassing traditional bit-and-crossbar slicing simulations.
- Developed the Non-Ideal PIM Accuracy (NIPA) evaluation model and its simulation framework, which leverages the unified error metric to rapidly assess relative accuracy and achieved a high correlation (up to 0.91) with ground-truth results.
- Implemented the complete non-slicing absolute accuracy evaluation method, which injects the unified error directly into model weights, achieving up to a $105.8\times$ speedup over conventional PIM simulators with an average evaluation error as low as 0.29%.

Simulation and Optimization of Digital–Analog Heterogeneous PIM-NoC (DAC'25)

May 2024—Nov. 2024

Tsinghua University | Advisor: Prof. Yu Wang

- Constructed a heterogeneous multi-core PIM-NoC simulation framework with flexible tile parameterization and configurable load ordering, providing an experimental platform for heterogeneous PIM system optimization.
- Authored the majority of the codebase, including over 4,000 lines of Python to streamline simulation workflows and data analysis.
- Designed an efficient search-based optimization algorithm on top of the simulator, achieving $2\text{--}3\times$ faster design-space exploration and up to 37.4% FoM improvement over homogeneous PIM-NoC baselines.

Network Partitioning for Heterogeneous Analog PIM–Digital Accelerators (ISQED'25)

Jan. 2024—May 2024

Tsinghua University | Advisor: Prof. Yu Wang

- Designed an automated exploration architecture for embedded analog in-memory computing–digital accelerator hybrid chiplet systems, optimized deployment across multiple accelerators and breakpoints, and performed neural network searches based on defined optimization objectives to develop near-optimal partitioning strategies.
- Developed a simulation model to validate different partitioning strategies, analyzed experimental data, and optimized computational load distribution to enhance system performance.
- Implemented multiple optimization techniques for rapid performance evaluation and refined code execution for efficiency, enabling up to 52% latency reduction with less than 1% accuracy loss in hybrid analog-digital PIM

accelerator systems.

TALKS

MIT Parallel Reading Group (Oct. 2025)
Tsinghua NICSEFC (Sep. 2025)
DAC Presentation (Jun. 2025)

AWARDS & RECOGNITION

Outstanding Graduate, Tsinghua University (Undergraduate) *2026*
Tsinghua University Comprehensive Scholarship (Top 2%, 2025)
Tsinghua University Comprehensive Scholarship (Top 5%, 2024)
Tsinghua University Comprehensive Scholarship (Top 8%, 2023)
1st Place, 2024 EE Dept Innovation Project Evaluation Annual Conference, Tsinghua

PROFESSIONAL SKILLS

Programming: C/C++, Python, Verilog, Java, Shell, LaTeX
Systems & Simulation: CUDA, MPI, OpenMP, Parlaylib, Gem5, MNSIM, BookSim, SPICE
Tools & Infrastructure: Docker, Git, GDB, perf, CMake
ML & Analysis: PyTorch, TensorFlow, QAT, MATLAB
Languages: English (TOEFL 107), Chinese (Native)